

Recurrent depth as an inference-time compute axis

Latent Loop Lab — one-page concept summary. DataForge 2026, Pathway Track.

The design pressure

A Transformer that reasons harder mostly reasons longer in text. Chain of thought buys accuracy by emitting intermediate tokens; each costs a forward pass, a slot in a key-value cache that grows with the sequence, and unparallelisable latency. The trace is legible, a real benefit, but the cost scales with how much thinking you want.

Recurrent latent computation takes the other route. One fixed-shape state is updated by the same parameters repeatedly and nothing is emitted until the end, so the budget is an iteration count R chosen at inference time. Memory stops growing with the amount of thinking, and the thinking stops being readable.

The claim

Repeated application of a **shared** state-update rule increases effective computational depth without generating intermediate language tokens — within the learned dynamics' capacity and stability limits.

Shared carries the argument. One block reused every step makes depth a loop bound at constant parameter count; a distinct block per step fixes depth when training ends. The artifact measures both: 11,169 parameters at any depth, against 39,585 with a hard ceiling at the trained depth of 4.

Mechanism, small enough to see

The task is directed-graph reachability. A learner moves R and watches a noisy-OR recurrence $h(r+1)[v] = \max(h(r)[v], 1 - \prod(1 - \alpha \cdot h(r)[u]))$ spread activation one edge per step. The invariant $h(R)[q] > \epsilon \iff d(s,q) \leq R$ is checked against an independent breadth-first search and holds on 10,000 seeded stratified cases, so R has an exact meaning: edges information can travel.

The same shape is then learned. A shared-weight recurrent graph network trains on distances 1–4 only, with R resampled from $\{1..4\}$ per batch, then runs at depths to 10 on distances to 10. Five seeds, frozen weights, only R varies:

R	learned	95% CI	distances ≤ 4	distances ≥ 5	exact
1	0.552	[0.546, 0.558]	0.621	0.506	0.550
4	0.700	[0.700, 0.700]	1.000	0.500	0.700
10	0.991	[0.972, 1.000]	0.994	0.988	1.000

Distances ≥ 5 sit at chance until R exceeds 4 — correct, since the answer is not yet knowable at the target — then climb to 0.988, solving paths longer than anything trained on by applying the same weights more times. Distances ≤ 4 peak at 1.000 near $R = 5$ and erode to 0.994 by $R = 10$ as the state norm saturates. Depth is not free.

Making this measure anything required fixing the data. An earlier generator built negatives as reversed chains, leaving the target with no incoming edges, so reachability collapsed to a one-hop lookup and accuracy was 1.000 everywhere. Each negative is now cut from a specific positive, matching node and edge counts exactly; two smaller gaps remain and are reported, not hidden.

Where BDH and BDH-CQ sit

BDH (Kosowski et al., arXiv:2509.26507) is a post-Transformer architecture of high-dimensional positive activations, low-rank communication and a recurrent associative state, with attention reformulated as synaptic memory; reported to rival GPT-2 at matched parameters from 10M to 1B, with sparse activations and synapse-level monosemanticity. Its GPU formulation combines ReLU low-rank transformations with linear attention and is not a Mamba-style state-space model.

BDH-CQ (Engdahl et al., arXiv:2608.09888) contains **two** recurrences that are routinely conflated. $S_t = U_\theta(S_{t-1}, D_t)$ is contextual memory advancing once per demonstration, driven by new input, with additive special case $S_t = S_{t-1} + U_\theta(D_t) \cdot H_{r+1} = F_\theta(H_r, S_K)$ is a query-time workspace advancing with no new input and S_K held fixed. Only the second is a reasoning step, and only the second is what this artifact models — the lab has no contextual recurrence at all. A live toy lets a learner drive each and produce the distinguishing failure: withhold a demonstration and no latent budget recovers the missing association.

	Chain of thought	Recurrent depth	BDH-CQ
Compute knob	tokens emitted	iterations R	latent effort level
State growth	KV cache grows	fixed shape	fixed-shape recurrent memory
Observability	readable trace	none	none

Evidence, labelled

BDH-CQ reports LOW 21%, MEDIUM 27%, HIGH 29.5% pass@2 on public ARC-AGI-1, at 22%, 11% and 0% cost reduction relative to HIGH; the headline is 29.5% at a computed \$0.00070 per task for 150M parameters, claimed to break the previously reported cost-accuracy Pareto frontier.

That is a **benchmark reported by the developers**, not a deployment. The report describes a black-box audit reproducing 29.5%, but by **co-authors** from Bielik and New York University — stronger than a self-report, weaker than an independent reproduction. Cost comparisons set measured hardware time for BDH-CQ against leaderboard costs for others that may be API prices, so they are not like-for-like. Dimensions and update rules are proprietary, so the mechanism is not externally auditable.

The most important open limitation is general: recurrent depth is reported to degrade past some depth in looped language models (Yang et al., arXiv:2605.26733), the same erosion this lab measures at toy scale. How far latent depth scales before that ceiling binds is unresolved.

Scope

Both models here are independent reimplementations written for teaching; neither is BDH or BDH-CQ, and no checkpoint of either was run. Every BDH and BDH-CQ figure is quoted and labelled paper-reported, never shown beside a measurement of ours. The learned model has ~11,000 parameters on one synthetic task, and graph reachability demonstrates computational depth, not semantic reasoning.

Continue with: the Dragon Hatchling paper; the BDH-CQ report; Geiping et al. (arXiv:2502.05171) on recurrent depth in a language model.